

1. Mr. Zadid's father had cancer. He knows that, even though cancer itself cannot be passed down from parents to children, there is 1% chance that the genetic change that increases the risk of cancer can be passed down (inherited). Mr. Zadid has 10^7 USD. He also knows that if is diagnosed with cancer, he will lose all of his money. 7 + 5
(CO3)
(PO3)

An insurance company comes to Mr. Zadid with a health insurance proposal. If he gives them an insurance premium of 5×10^6 USD, they will reimburse him 9×10^6 USD ONLY IF he gets diagnosed with cancer. Mr. Zadid's utility function is $U(x) = \ln(1 + x)$.

- a) Assume that you are a rational agent. Recommend whether Mr. Zadid should buy the health insurance or not.

Solution:

The expected utility for Mr. Zadid if he does not buy the insurance is:

$$0.99 \times \ln(1 + 10^7) + 0.01 \times \ln(1 + 0) \approx 15.957$$

When taking the insurance, the expected utility for Mr. Zadid is:

$$0.99 \times \ln(1 + (5 \times 10^6)) + 0.01 \times \ln(1 + (5 \times 10^6) + (9 \times 10^6) - 10^7) \approx 15.423$$

Since the expected utility is lesser, Mr. Zadid should not buy the insurance.

Rubric:

- 2 points each for expected utility
- 3 points for decision

- b) Comment on why the insurance company is offering such a deal where they have a chance to lose 4×10^6 USD.

Solution:

The expected monetary value for the insurance is:

$$0.99 \times (5 \times 10^6) + 0.01 \times ((5 \times 10^6) - (9 \times 10^6)) = 4.91 \times 10^6 \text{ USD.}$$

Since the value is positive, the insurance company can offer the same insurance to as many people similar to Mr. Zadid as possible and gain a net profit.

Rubric:

- 2 points for the expected monetary value
- 3 points for explanation

2. A meme is an image, video, piece of text, etc., typically humorous in nature, that is copied and spread rapidly by internet users, often with slight variations. 3

Create a meme related to this course. A text description will suffice.

Solution:

**PEOPLE: AI IS GOING TO KILL ALL HUMANS.
AI:**

